

Sakshi Joshi

PhD Research Scholar, IIT Madras | AI4Bharat

@ da23d201@smail.iitm.ac.in in LinkedIn GitHub Google Scholar
619, New Academic Complex-2, IIT Madras, TN, India

Education

Ongoing July 2023	Indian Institute of Technology (IIT), Madras PhD, Data Science and Artificial Intelligence CGPA: 8.2/10 Advisor : Prof. Mitesh M. Khapra	Chennai, India
Jul 2021 Jun 2017	Medicaps University B.Tech, Computer Science & Engineering CGPA: 8.2/10	Indore, India

Experience

Present Jan 2023	AI4Bharat, IIT Madras [🔗] PhD Research Scholar — Automatic Speech Recognition / Advisor: Prof. Mitesh M. Khapra Working on automatic speech recognition for low-resource domains and underrepresented demographics by building datasets, benchmarks, and models, including SRUTI, IndicContextEval, and child-centric ASR systems for educational settings. Project Associate — Automatic Speech Recognition / Advisor: Prof. Mitesh M. Khapra Built data pipelines and conducted evaluation for large-scale multilingual ASR benchmarks and datasets across Indian languages, contributing to projects such as LAHAJA, Svarah, and IndicVoices.	Chennai, India July 2023 – Present
Dec 2022 Jul 2022	CEET, IIT Madras Research Park [🔗] Project Associate — Machine Learning / Advisor: Prof. Ashok Jhunjhunwala Collaborated with Uniphore on voice-based applications and contributed to BuildClub AI courses for school students.	Chennai, India
May 2021 Feb 2021	STARC Lab, IIT Indore [🔗] Data Science and Research Intern / Advisor: Dr. Abhirup Datta Built pipelines to process large-scale GNSS (GPS and NavIC) receiver data for ionospheric analysis and performed statistical correlation with space weather parameters for geomagnetic storm characterization.	Remote
Dec 2020 May 2020	Codenscious AI [🔗] Machine Learning Intern Worked on financial data analysis and development of custom machine learning algorithms as part of an industry-focused ML and deep learning training program.	Indore, India

Publications

C=Conference, P=Preprint, *=Equal Contribution

- [P.1] **IndicContextEval: A Benchmark for Evaluating Context Utilisation in Audio Large Language Models Across 8 Indic Languages**
Sakshi Joshi*, Dhruv Rathee*, Sanskar, Eldho Ittan George, RJ Hari, Kaushal Bhogale, Mitesh M. Khapra
Under Review [Interspeech 2026]
- [C.1] **Recognizing Every Voice: Towards Inclusive ASR for Rural Bhojpuri Women** [🔗]
Sakshi Joshi, Eldho Ittan George, Tahir Javed, Kaushal Bhogale, Nikhil Narasimhan, Mitesh M. Khapra
Annual Conference of the International Speech Communication Association
🏆 Best Theme Paper Award [Interspeech 2025]
- [C.2] **Towards Bringing Parity in Pretraining Datasets for Low-resource Indian Languages** [🔗]
Kaushal Santosh Bhogale, Devrat Mehendale, Tahir Javed, Devbrat Anuragi, Sakshi Joshi, Sai Sundaresan, Aparna Ananthanarayanan, Sharmistha Dey, Sathish Kumar Reddy G, Anusha Srinivasan, Abhigyan Raman, Pratyush Kumar, Mitesh M. Khapra
IEEE International Conference on Acoustics, Speech and Signal Processing [ICASSP 2025]

- [C.3] **LAHAJA: A Robust Multi-accent Benchmark for Evaluating Hindi ASR Systems** [🔗]
Tahir Javed, Janki Nawale, [Sakshi Joshi](#), Eldho George, Kaushal Bhogale, Deovrat Mehendale, Mitesh M. Khapra
Annual Conference of the International Speech Communication Association [Interspeech 2024]
- [C.4] **IndicVoices: Towards Building an Inclusive Multilingual Speech Dataset for Indian Languages** [🔗]
Tahir Javed, Janki Atul Nawale, Eldho Ittan George, [Sakshi Joshi](#), Kaushal Santosh Bhogale, Deovrat Mehendale, Ishvinder Virender Sethi, Aparna Ananthanarayanan, Pratyush Kumar, Mitesh M. Khapra
62nd Annual Meeting of the Association for Computational Linguistics [ACL 2024]
- [C.5] **Svarah: Evaluating English ASR Systems on Indian Accents** [🔗]
Tahir Javed, [Sakshi Joshi](#), Vignesh Nagarajan, Sai Sundaresan, Janki Nawale, Abhigyan Raman, Kaushal Bhogale, Pratyush Kumar, Mitesh M. Khapra
Annual Conference of the International Speech Communication Association [Interspeech 2023]

Projects

- ASR for Rural Women — Recognizing Every Voice** [Data] [Code] Jul 2024 – Aug 2025
- > Built SRUTI, a benchmark dataset of rural Bhojpuri women's speech across domains such as healthcare, agriculture, governance, and finance, highlighting severe performance gaps in existing ASR systems.
 - > Proposed a scalable synthetic data generation pipeline using 25–30 seconds of speech per speaker to create 100+ hours of data, improving ASR performance by 4.7 WER in low-resource settings.
- IndicContextEval — Contextual ASR Benchmarking** [Under Review] Aug 2025 – Present
- > Developed a 56-hour multilingual benchmark spanning 8 Indian languages and 23 domains to evaluate contextual grounding in AudioLLMs under controlled prompting settings.
 - > Designed a 7-level prompting framework to systematically analyze how models utilise domain metadata, natural language descriptions, and entity-level context, revealing significant variation in contextual reasoning.
- Automatic Assessment of Oral Reading Fluency** [Ongoing] Jan 2026 – Present
- > Building a large-scale multilingual child speech corpus for oral reading fluency across Indian languages, capturing grade-level reading tasks and detailed reading errors.
 - > Designing AudioLLM-based models to jointly assess accuracy, fluency, and prosody, along with evaluation pipelines using WER, PER, and miscue-based metrics for scalable literacy assessment.
- Child Speech Recognition for Educational Settings** [Ongoing] Jan 2026 – Present
- > Building multi-turn conversational speech datasets capturing teacher-student interactions across subjects, incorporating dialogue context into ASR.
 - > Designing context-aware ASR models conditioned on conversational history and evaluating them using context-sensitive metrics such as Named Entity Error Rate (NEER).

Awards and Achievements

- Best Theme Paper Award at Interspeech 2025** Recognizing Every Voice: Towards Inclusive ASR for Rural Bhojpuri Women
- 2nd Prize — Poster Presentation, WSAI Annual Research Showcase 2025** Won second prize at the Wadhvani School of AI Annual Research Showcase; awarded travel grant covering Interspeech 2025 registration fees
- Google DeepMind** Selected to attend the Google DeepMind Research Symposium 2025
- Travel Grant — IndoML 2023** Awarded travel support to attend IndoML 2024
- GATE CS** Qualified with 96.33 percentile (2023) and 94.33 percentile (2022)
- Medhavi Vidyarthi Protsahan Scholarship** Awarded for academic excellence in 10+2 examination